

WATER-QUALITY HAZARD

Drinking Water Contamination

Water can meet systemwide standards yet still be affected by a local service line, household plumbing, private well, or temporary treatment failure.



1 Understand the factor

What it is - and why it matters

Drinking-water contamination can originate at the source, treatment plant, distribution system, service line, household plumbing, storage tank, or private well. Potential contaminants include microbes, lead, arsenic, nitrate, PFAS, solvents, pesticides, disinfection byproducts, and naturally occurring minerals. Health significance depends on the contaminant, concentration, duration, route, and the person exposed. A water-quality report describes a system, not necessarily the exact water from a particular faucet at every moment.

How risk can reach a household

- Source-water pollution or treatment failure can introduce microbial or chemical contaminants across a service area.
- Corrosion can release lead or copper from service lines, solder, fixtures, and building plumbing, especially after water-chemistry changes.
- Groundwater plumes, septic systems, fertilizers, or naturally occurring minerals can affect private wells without obvious taste or odor.
- Main breaks, low pressure, cross-connections, flooding, or premise plumbing can create localized contamination events.

What people often miss

- Clear, good-tasting water can still contain lead, nitrate, arsenic, PFAS, or microbes; appearance is not a safety test.
- A utility's Consumer Confidence Report uses system samples and regulatory schedules; it may not capture a home's service line or intermittent issue.
- Boiling kills many microbes but can concentrate some chemicals and does not remove lead or PFAS.
- Filters work only for contaminants listed on their certification and must be installed, operated, and replaced correctly.

? Five checks before buying

1

Identify the water provider, source type, treatment, service-line material, pressure zone, and whether the home has a private well or storage system.

2

Read recent Consumer Confidence Reports, violation notices, source-water assessments, boil advisories, and local lead-service-line inventory.

3

Use a state-certified laboratory for targeted testing based on local geology, plumbing age, nearby releases, and household vulnerability.

4

Inspect the wellhead, setback, casing, cap, treatment equipment, softener, filters, and plumbing; obtain maintenance and sampling records.

5

Ask about prior contamination, bottled-water use, filter systems, plumbing replacement, water claims, and any recorded well restrictions.

IMPORTANT DISTINCTION

Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 15

H Health impacts

- Microbial contamination can cause gastrointestinal illness and can be severe for infants, older adults, pregnant people, and immunocompromised residents.
- Skin contact and inhalation may matter for certain volatile chemicals, while other contaminants are mainly an ingestion concern.
- Lead can harm neurological development; nitrate can be dangerous for infants; arsenic and some organic chemicals can raise long-term disease risks.
- Uncertainty about daily water safety can affect hydration, cooking, infant feeding, costs, and stress.

P Property impacts

- Contamination may require service-line replacement, plumbing changes, a new well, treatment equipment, bottled water, or connection to public water.
- Known well or plumbing problems can affect inspection, appraisal, lender requirements, disclosure, and resale.
- Filters and treatment systems add ongoing testing, maintenance, electricity, media replacement, and disposal costs.
- Leaks and corrosive water can damage fixtures, appliances, pipes, water heaters, and finishes in addition to health concerns.

I Insurance implications

- Homeowners insurance generally does not cover the cost of correcting gradual water contamination, replacing an old service line, or treating a private well.
- Service-line endorsements may cover certain physical failures but do not necessarily cover contamination or code-driven replacement.
- Sudden accidental plumbing damage may be covered while the contaminant, testing, loss of water use, and preventive upgrades are excluded or limited.
- Public relief programs, responsible-party claims, or utility assistance may exist, but eligibility and timing should not be assumed.

Possible long-term health implications of buying nearby

- Long-term health risk depends on repeated dose, so the most protective step is identifying the contaminant and stopping exposure rather than relying on symptoms.
- A private well can change over time after flooding, drought, nearby construction, septic failure, or plume movement, requiring periodic retesting.
- Infants, pregnant people, children, and people with kidney or immune conditions may need lower thresholds or faster action for certain contaminants.
- A properly selected certified treatment system can reduce risk, but performance must be confirmed with post-treatment sampling.

Evidence lens

Health effects for regulated contaminants are supported by substantial evidence, but a home's risk requires representative sampling, a certified laboratory, and interpretation against current health and regulatory guidance.

Plan more carefully for

Children

Older adults

Pregnancy

Heart/lung conditions

Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

FACTOR 15

V Verify the actual risk

- Review utility reports and test the home's water when plumbing, well, local contamination, or vulnerable occupants justify it.
- Sample correctly at the point and time recommended by the laboratory; use first-draw, flushed, raw-well, or treated samples as appropriate.

P Protect the property

- Replace lead-containing service lines and fixtures when indicated; protect wells from surface drainage, pests, and chemical storage.
- Install only treatment certified for the identified contaminant and size it for actual flow and concentration.

M Monitor and respond

- Follow boil, do-not-drink, or do-not-use advisories exactly and use safe alternate water for formula, cooking, medications, and pets.
- Retest after remediation, flooding, well repair, treatment changes, or a nearby contamination notice.

I Prepare insurance records

- Clarify service-line, plumbing, water-damage, mold, pollution, testing, and loss-of-use coverage.
- Keep laboratory reports, chain-of-custody forms, filter certifications, maintenance logs, receipts, and agency notices.

REAL-LIFE EXAMPLE

Flint Drinking Water Crisis

Beginning in 2014

After Flint, Michigan, changed its water source without adequate corrosion control, lead leached from pipes and service lines into drinking water. Residents also faced concerns about water quality and disease. Federal, state, and local responses included emergency measures, testing, bottled water and filters, infrastructure work, enforcement, and long-term recovery.

Why it matters: Flint demonstrated that source changes, corrosion chemistry, aging plumbing, sampling practices, communication, and delayed response can turn a system decision into a household exposure and trust crisis.

[Open official incident record - epa.gov >](#)

Questions to resolve before making a decision

1. Identify the water provider, source type, treatment, service-line material, pressure zone, and whether the home has a private well or storage system.
2. Read recent Consumer Confidence Reports, violation notices, source-water assessments, boil advisories, and local lead-service-line inventory.
3. Use a state-certified laboratory for targeted testing based on local geology, plumbing age, nearby releases, and household vulnerability.
4. Inspect the wellhead, setback, casing, cap, treatment equipment, softener, filters, and plumbing; obtain maintenance and sampling records.

Official sources and mapping tools

[EPA Flint drinking-water information - epa.gov >](#)[EPA Consumer Confidence Reports - epa.gov >](#)[EPA private drinking-water wells - epa.gov >](#)[EPA certified drinking-water laboratories - epa.gov >](#)